

Numerical Methods for Engineers

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Fifth Edition

Mathematical Modeling and Engineering Problem solving

Chapter 1

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- Numerical methods are techniques by which mathematical problems are formulated so that they can be solved with arithmetic operations (involves large numbers of tedious arithmetic calculations).

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Non-Computer Methods

1. Analytical (Exact) Methods: can be derived for a limited class of problems (e.g. linear models with a simple geometry). Most real problems are nonlinear and involve complex shapes and processes.
2. Graphical Solutions: (e.g. Plots) Limited to problems that can be described using **three** or **fewer** dimensions.
3. Calculators: Slow, tedious

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- Although analytical solutions are still extremely valuable for problem solving and for providing insight, numerical methods represent alternative that greatly enlarge our capabilities to solve problems.

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Why study numerical methods?

1. Numerical methods are extremely powerful problem solving tools (can handle **large systems of equations**, **nonlinearities** etc.)
2. Numerical methods are an efficient vehicle for learning to program:
   - Illustrate the power and limitations of computers
   - Acknowledge and control errors of approximation.
3. Numerical methods provide a vehicle to reinforce your understanding of mathematics **by reducing higher mathematics to basic arithmetic equations.**

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(a) *Part 2:* Roots of equations
Solve $f(x) = 0$ for x.

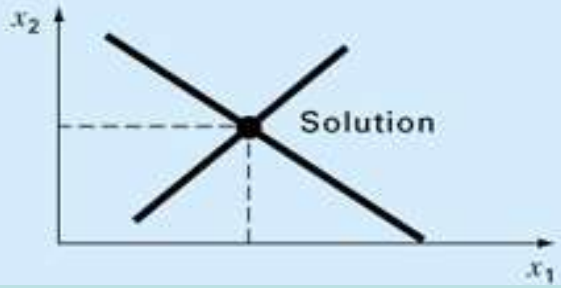


Every part in this book requires some mathematical background

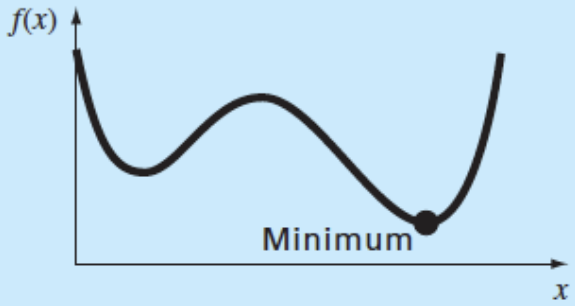
(b) *Part 3:* Linear algebraic equations
Given the a's and the c's, solve

$$a_{11}x_1 + a_{12}x_2 = c_1$$
$$a_{21}x_1 + a_{22}x_2 = c_2$$

for the x's.

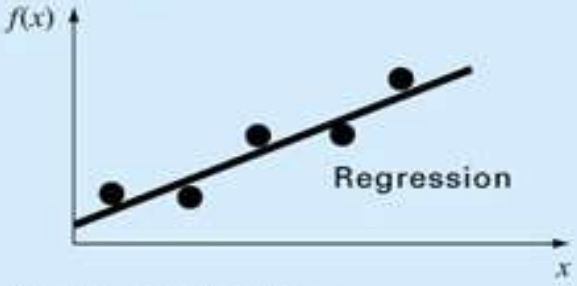


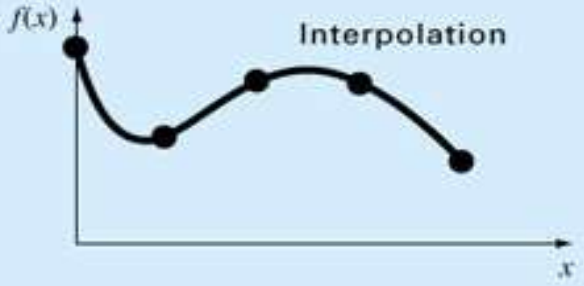
(c) *Part 4:* Optimization
Determine x that gives optimum $f(x)$.



-- Single and multi-variable unconstrained optimization
-- Constrained optimization with emphasis on linear programming.

(d) *Part 5:* Curve fitting



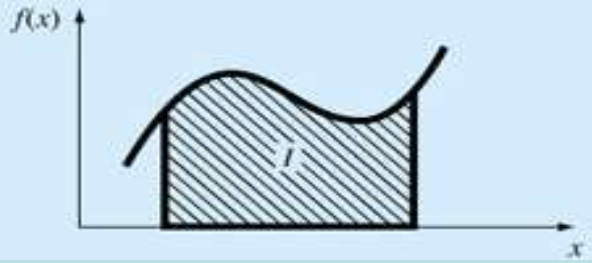


- **Regression**: There is error in data
- **Interpolation**: Determine intermediate values for error-free data.

(e) *Part 6:* Integration

$I = \int_a^b f(x)\,dx$

Find the area under the curve.



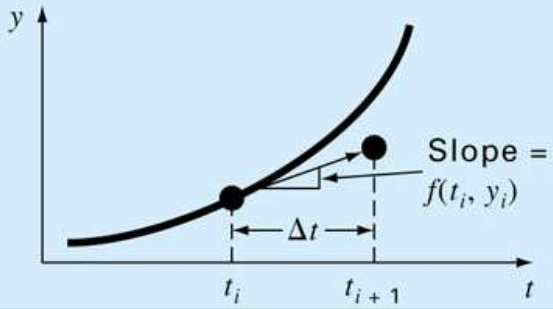
(f) *Part 7:* Ordinary differential equations

Given

$$\frac{dy}{dt} \simeq \frac{\Delta y}{\Delta t} = f(t, y)$$

solve for y as a function of t.

$$y_{i+1} = y_i + f(t_i, y_i)\,\Delta t$$



- Initial value problems

- Boundary value problems

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Computers are great tools, however, without fundamental understanding of engineering problems, they will be useless.

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Engineering Simulations

Finite element analysis (FEA) and product design services
Computational Fluid Dynamics (CFD)

Molecular Dynamics
Particle Physics
N-Body Simulations

Earthquake simulations

Development of new products and performance improvement of existing products

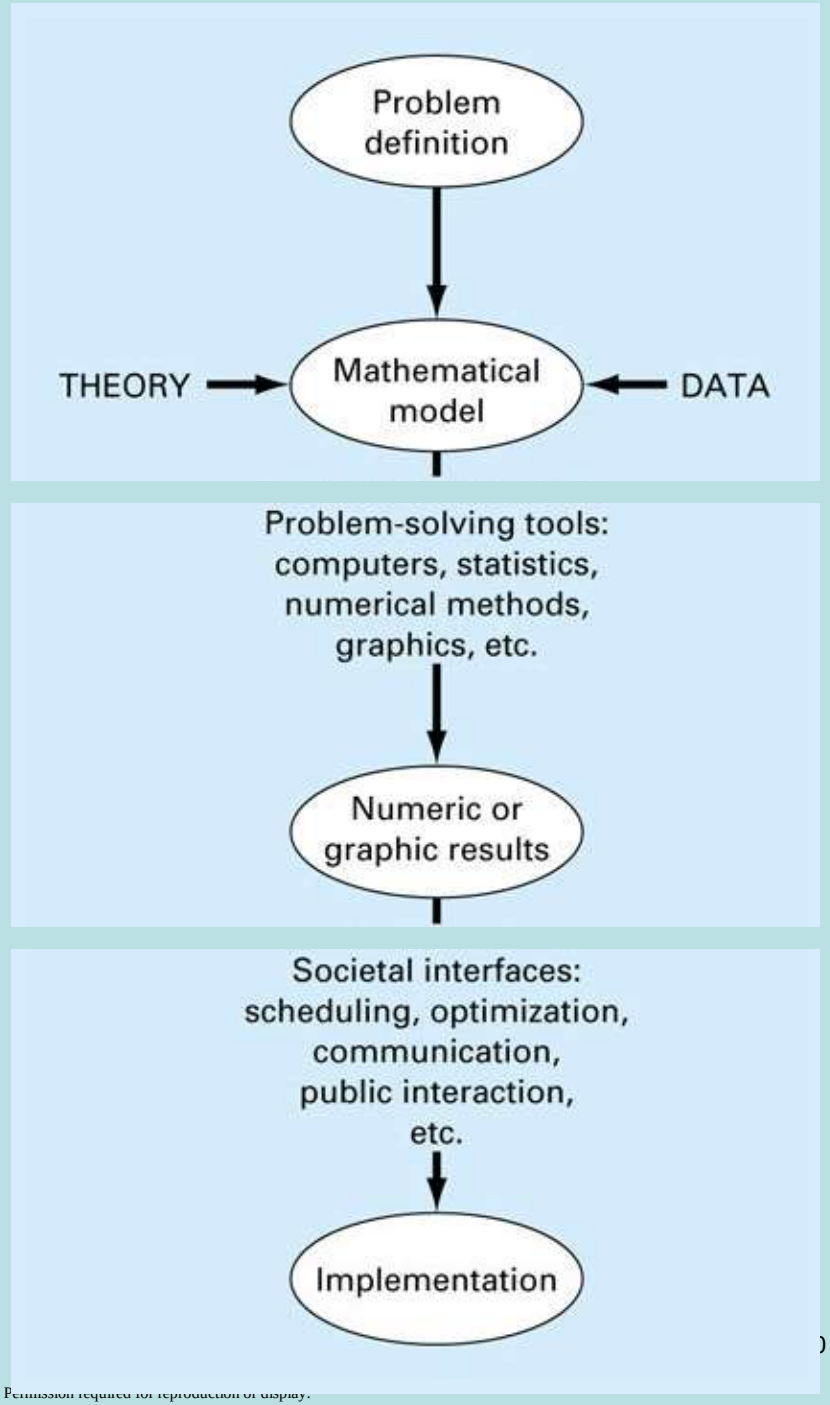
Benefits of Simulations

Cost savings by minimizing material usage.
Increased speed to market through reduced product development time.

Optimized structural performance with thorough analysis
Eliminate expensive trial-and-error.

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The Engineering Problem Solving Process



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Mathematical Model

- A formulation or equation that expresses the essential features of a physical system or process in mathematical terms.

  dependent variable = f(independent variables, parameters,forcing functions)

- **Dependent variable**: reflects the **behaviour** or **state** of the system.
- **Independent variable**: generally time or space.
- **Parameters**: systems' properties.
- **Forcing functions**: external influences

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Parachutist Problem

- Problem statement: A parachutist of mass 68.1kg jumps out of a stationary hot air balloon. We would like to compute his **velocity** prior to opening the chute.

- First let's solve it Analytically!

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Newton's 2nd law of Motion

- *"The time rate change of momentum of a body is equal to the resulting force acting on it."*

- Formulated as **F = m.a**

  F = net force acting on the body

  m = mass of the object (kg)

  a = its acceleration (m/s^2)

- Some complex models may require more sophisticated mathematical techniques than simple algebra
  - Example, modeling of a falling parachutist:

$$F = F_D + F_U$$

F_U = Force due to air resistance = $-cv$ (***c* = *drag coefficient***)

Accounts for the shape or surface roughness of the object, that affect air resistance.

F_D = Force due to gravity = mg



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$$a = \frac{F}{m}$$

a: dependent variable reflecting systems' behaviour

F: forcing function

m: parameter representing the property of the system

- No independent variable since we are not predicting how acceleration varies with time.
- **Goal**: Use Newton's second law to determine the terminal velocity of a free-falling body near Earth's

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$$\frac{dv}{dt} = \frac{F}{m}$$

$$F = F_D + F_U$$

$$F_D = mg$$

$$F_U = -cv$$

$$\frac{dv}{dt} = \frac{mg - cv}{m}$$

- This is a first order ordinary **differential equation.** We would like to solve for v (velocity).
- It can **not** be solved using algebraic manipulation
- Analytical Solution:

  If the parachutist is initially at rest (v=0 at t=0), using calculus dv/dt can be solved to give the result:

$$\frac{dv}{dt} = g - \frac{c}{m}v$$

Dependent variable

Independent variable

$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right)$$

Forcing function

Parameters

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Analytical Solution

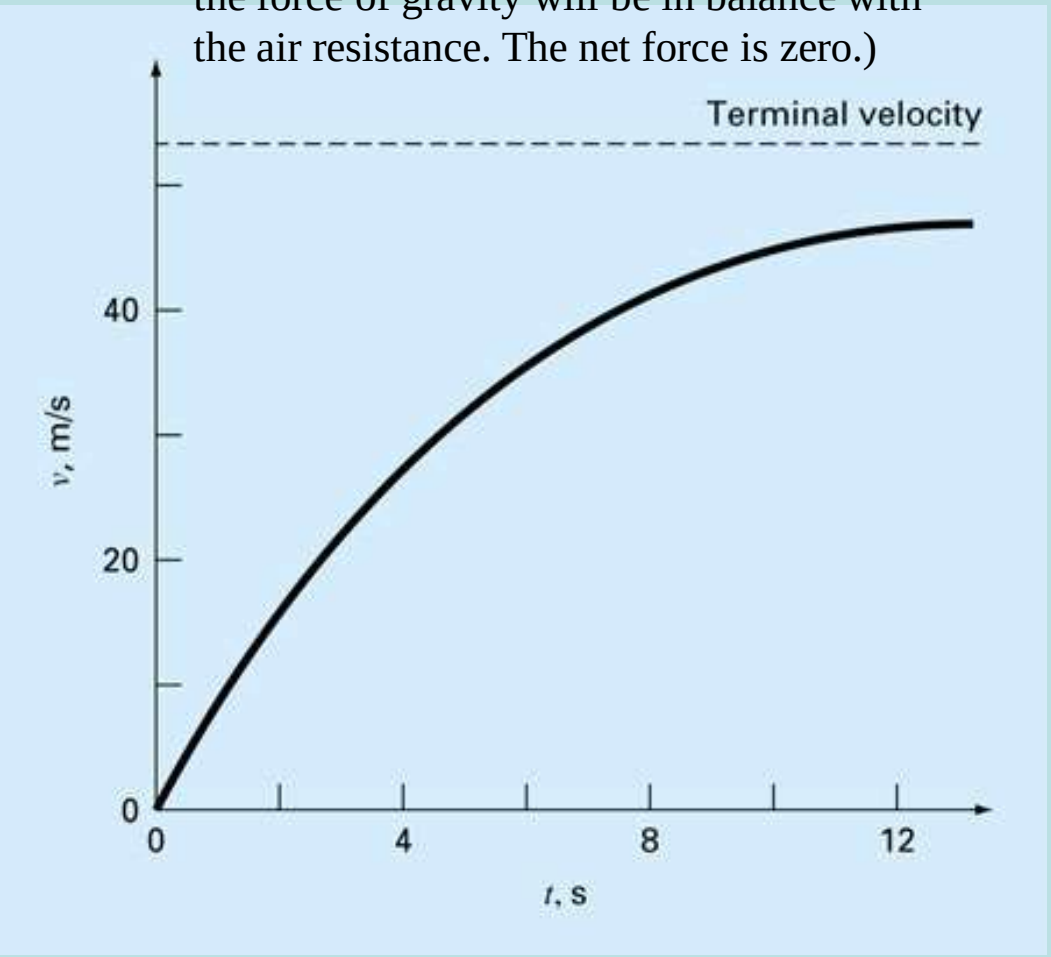
$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right)$$

If ***v(t)*** could not be solved **analytically**, then we need to use a numerical method to solve it

g = 9.8 m/s^2 c =12.5 kg/s
m = 68.1 kg

t (sec.)	V (m/s)
0	0
2	16.40
4	27.77
8	41.10
10	44.87
12	47.49
∞	53.39

Terminal velocity is constant (eventually the force of gravity will be in balance with the air resistance. The net force is zero.)



Run ***analpara.m, analpara2.m*, and ***analpara4.m*** at W:\228\MATLAB\1-2

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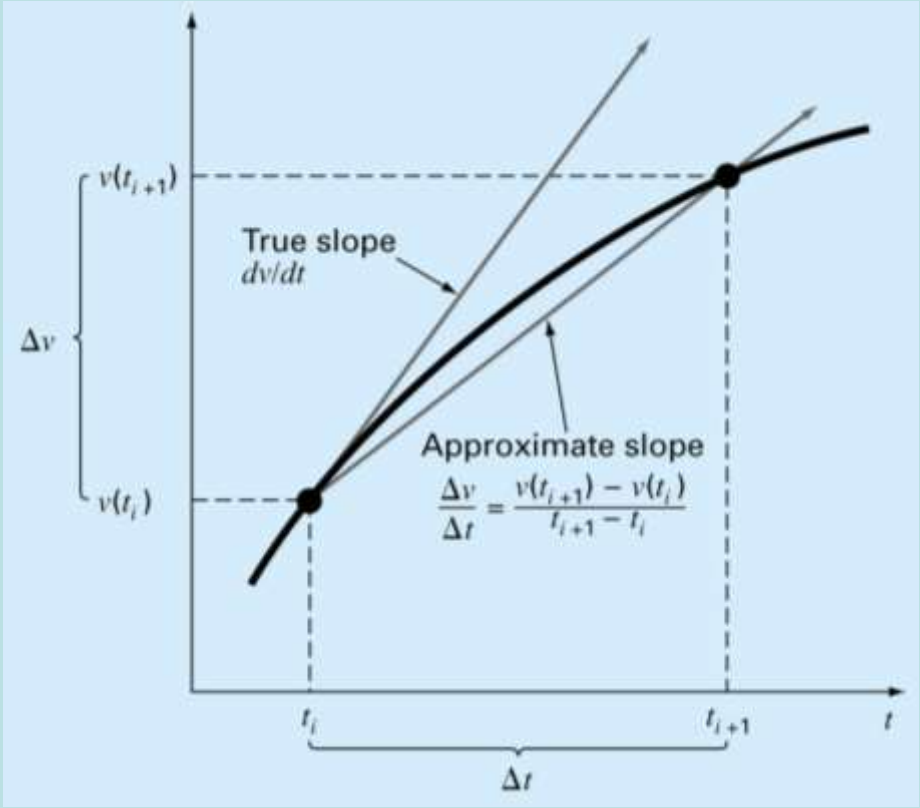
Numerical Solution

$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} \quad \text{.....} \quad \frac{dv}{dt} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$

$$\frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = g - \frac{c}{m} v(t_i)$$

This equation can be rearranged to yield

$$v(t_{i+1}) = v(t_i) + [g - \frac{c}{m} v(t_i)](t_{i+1} - t_i)$$

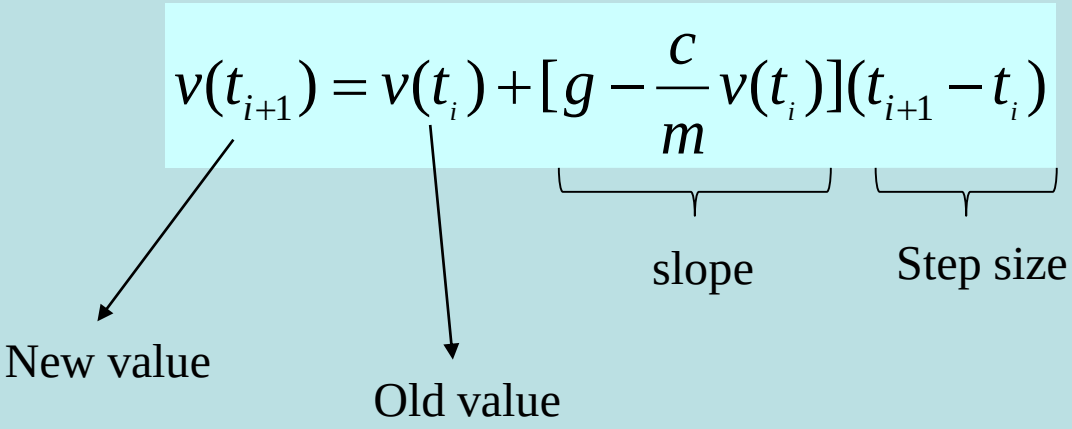


t (sec.)	V (m/s)
0	0
2	19.60
4	32.00
8	44.82
10	47.97
12	49.96
∞	53.39

$\Delta t = 2$ sec

To minimize the error, use a smaller step size, Δt
No problem, if you use a computer!

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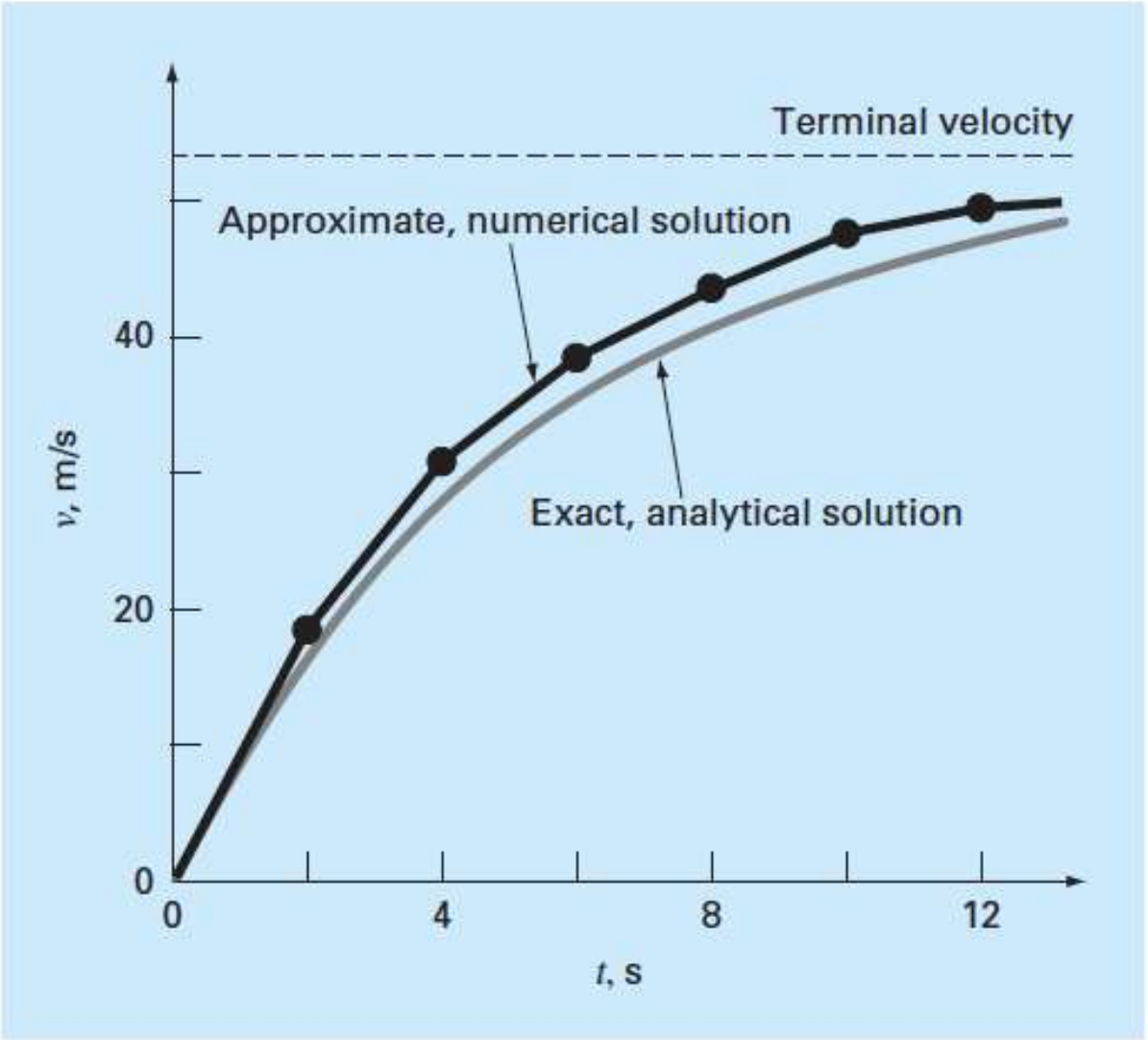


FIGURE 1.5
Comparison of the numerical and analytical solutions for the falling parachutist problem.

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Analytical vs. Numerical solution

m=68.1 kg c=12.5 kg/s
g=9.8 m/s

t (sec.)	V (m/s)
0	0
2	16.40
4	27.77
8	41.10
10	44.87
12	47.49
∞	53.39

Δt = 2 sec

t (sec.)	V (m/s)
0	0
2	19.60
4	32.00
8	44.82
10	47.97
12	49.96
∞	53.39

Δt = 0.5 sec

t (sec.)	V (m/s)
0	0
2	17.06
4	28.67
8	41.95
10	45.60
12	48.09
∞	53.39

Δt = 0.01 sec

t (sec.)	V (m/s)
0	0
2	16.41
4	27.83
8	41.13
10	44.90
12	47.51
∞	53.39

$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right)$$

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m}v(t_i)\right]\Delta t$$

*Run ***numpara2.m*** at W:\228\MATLAB\1-2

CONCLUSION: If you want to minimize the error, use a smaller step size, Δt

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